

universal connectivity, converged information technology (IT) and operational technology (OT) networks

- *Procurement.* Supplier visibility, track and trace
- *Maintenance and reliability.* Asset health monitoring, predictive maintenance, field workforce effectiveness, environment, health and safety management
- *Manufacturing operations.* Shop-floor workforce flexibility and efficiency, automated touchless factory quality assurance, compliance and analytics, energy monitoring and management, real-time operational intelligence

Bridging the gap between machines and IT

Technologies that can bridge the gap between OT and IT are smart assets, multi-disciplined control and information, secure network infrastructure, mobility and scalable computing. **Information-driven manufacturing leads to ease of use, better quality control, increased safety, increased productivity, improved design through simulation (CAD/CAM), energy savings, reduced**

labour costs, reduced or removed routine manual and clerical tasks, and so on. It requires building a collaborative team, reliable and secure infrastructure, technology upgrade for the digital world, developing skills (educating and training employees), partnerships to close the gaps and solve challenges for successful digital transformation.

Between OT and IT, there lie various manufacturing operations and management processes that help achieve a connected enterprise. OT includes materials and transport, machines and testers, controllers and scanners, and 3D printers. Whereas, IT includes ERP, financials, HR, logistics, quality and warehouse.

Smart in-line inspection systems and plant monitoring systems. Defining a problem gives half its solution. Utilising resources effectively, capitalising human resources intelligently, managing finances and deploying the right technology can solve any business problem. **Trending technologies applied in Industry 4.0 are the Industrial IoT (IIoT), RFID, AR and virtual reality (VR) for enhanced visualisation**

and learning, cyber physical systems, computer vision, 3D printing with multi-material capability, automated guided vehicles, private LTE-enabled connectivity, cobots, smart sensors, network security, implementation of OPC Unified Architecture (OPC UA, a machine-to-machine communication protocol for industrial automation), actuators and controllers, additive manufacturing, digital twins, wireless instrumentation, cybersecurity and industrial design.

A cyber physical assembly line includes the following smart machine interactions: voice interaction, WhatsApp messaging, voice calling, augmented experience using AI, machine learning, NLP and analytics, remote access to human-machine interface (HMI).

Maintenance and electrical fault finding can be achieved in four steps, as explained below.

1. Design a CAD model, which is a digital replica of a physical system.
2. Create an electrical schematic design. When synced with the CAD model, add electro-mech layer of information.

Pulsonix 3D PCB Design Software

PCB design done right

Advanced 3D PCB Design Features

- Co-design in integrated PCB / 3D environment
- Place and move components in 3D
- Automatic synchronisation of electrical and mechanical design
- Real-time 3D DRC for clash detection
- Realistic visualisation of the PCB
- Visualise internal layers with exploded view
- Measure between objects in 3D

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Modern 3D Design and View

Mechanical fit and placement errors are a thing of the past with Pulsonix 3D PCB design. Real-time 3D DRC finds clashes during design, and you can easily fix them with either 2D PCB or 3D editing.

That's PCB design done right!

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